From RSVP Fields to Mechanistic Self-Aware Machines

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Abstract

This paper aims to unify field-theoretic physics with cognitive modeling by integrating the Relativistic Scalar-Vector Plenum (RSVP) framework with Bennett's mechanistic model of AI self-awareness. Using category theory, we define **Mech** and **RSVP**, construct a functor $F : \mathbf{Mech} \to \mathbf{RSVP}$, and prove properties including non-diverging Grüneisen parameters via q-entropy and sheaf cohomology, incorporating error-correction inspired by magnetic proofreading self-assembly. Results include rigorous categorical constructions, functoriality proofs, and computational recipes, with self-awareness modeled as a homotopy fixed point. Implications suggest a unified framework for physics, cognition, and self-assembly, with testable predictions for AI and physical systems.

1 Introduction

1.1 Prerequisites

Readers should be familiar with category theory (objects, morphisms, functors, natural transformations [1, 8]), differential geometry (smooth manifolds, tangent bundles [7]), and sheaf theory (presheaves, gluing conditions [10, 3]). Knowledge of statistical physics (entropy, critical phenomena [16]) and causal inference (structural causal models [13]) is assumed.

1.2 Motivation and Overview

The Relativistic Scalar-Vector Plenum (RSVP) theory models physical systems as scalar (Φ), vector ($\sqsubseteq$), and entropy ($\mathcal{S}_q$) fields over a fibered category, addressing critical phenomena without divergences using q-entropy and sheaf cohomology [14, 16, **?**]. Bennett's mechanistic model posits that self-awareness arises from causal inference and intent attribution, implemented computationally [2]. Magnetic proofreading self-assembly demonstrates local error correction to achieve high-fidelity structures, offering a physical analogy to RSVP entropy control [9]. This paper constructs a functor $F : \mathbf{Mech} \to \mathbf{RSVP}$, mapping mechanistic operations to field dynamics, and integrates proofreading mechanisms to stabilize entropy.

This work unifies thermodynamic criticality, cognitive modeling, and physical self-assembly. The exposition proceeds with preliminaries, category definitions, functorial mappings, hardware-type mappings, computational implementation, worked examples, and philosophical interpretations, concluding with testable predictions. TikZ diagrams illustrate key concepts.

2 Mathematical Preliminaries

2.1 Category Theory

A category $\mathcal{C}$ consists of objects $\mathrm{Ob}(\mathcal{C})$, morphisms $\mathrm{Hom}_{\mathcal{C}}(A, B)$, associative composition $\circ$, and identities id_A [1, 8]. A functor $F : \mathcal{C} \to \mathcal{D}$ preserves composition and identities. A natural transformation $\eta : F \Rightarrow G$ assigns morphisms $\eta_A : F(A) \to G(A)$ such that $G(f) \circ \eta_A = \eta_B \circ F(f)$. An adjunction $F \dashv G$ involves unit η and counit ϵ satisfying triangle identities [15].

2.2 Sheaf Theory

A sheaf $\mathcal{F}$ on a topological space X assigns data $\mathcal{F}(U)$ to open sets $U \subseteq X$, with restriction maps and gluing: for cover $\{U_i\}$, compatible sections $\{s_i \in \mathcal{F}(U_i)\}$ glue uniquely [10, 3]. Cohomology $\check{H}^n(X, \mathcal{F})$ measures gluing obstructions.

2.3 Entropy and Criticality

Boltzmann-Gibbs entropy $S = -k \sum p_i \log p_i$ fails at critical points. Tsallis q-entropy $S_q = k \frac{1-\sum p_i^q}{q-1}$ restores extensivity [16]. The Grüneisen parameter $\Gamma = \frac{\alpha}{C_p}$ quantifies anharmonicity, non-diverging with q-entropy [**?**, 14].

2.4 Mechanistic AI and Self-Assembly

Structural causal models (SCMs) define variables and equations for inference [13]. Self-awareness involves intent attribution and self-modeling [2, 6]. Magnetic proofreading self-assembly uses external fields to stabilize correct configurations, analogous to entropy control [9].

3 RSVP Theory Formalism

3.1 Prerequisites

Assume a smooth manifold $\mathcal{M}$ with tangent bundle $T\mathcal{M}$. Sheaves are on the site of open sets with Zariski topology, valued in **Ab** (abelian groups). Fields satisfy PDEs derived from action functionals [7].

Definition 3.1. Objects: Triples $(\Phi, \sqsubseteq, \mathcal{S}_q)$ where: - $\Phi \in C^\infty(\mathcal{M}, \mathbb{R})$ is a scalar field (e.g., $\Phi(x) = x^2$ for a quadratic potential). - $\sqsubseteq \in \Gamma(T\mathcal{M})$ is a vector field (e.g., $\sqsubseteq = \partial_x$ on $\mathbb{R}$). - $\mathcal{S}_q$ is a sheaf of entropies valued in $\mathbb{R}$, satisfying gluing [10]. Morphisms: $f : (\Phi, \sqsubseteq, \mathcal{S}_q) \to (\Phi', \sqsubseteq', \mathcal{S}'_q)$ as triples $(f_\Phi, f_\sqsubseteq, f_{\mathcal{S}_q})$, with f_Φ smooth (e.g., restriction $\Phi \to \Phi|_U$), $f_\sqsubseteq$ a bundle morphism, $f_{\mathcal{S}_q}$ a sheaf morphism. Composition: $(g \circ f) = (g_\Phi \circ f_\Phi, g_\sqsubseteq \circ f_\sqsubseteq, g_{\mathcal{S}_q} \circ f_{\mathcal{S}_q})$. Identities: $\mathrm{id}_{(\Phi, \sqsubseteq, \mathcal{S}_q)} = (\mathrm{id}_\Phi, \mathrm{id}_\sqsubseteq, \mathrm{id}_{\mathcal{S}_q})$. **RSVP** is enriched over $\mathbf{Vect}_\mathbb{R}$[1] and **Top**[2], with monoidal product $\otimes$ as field concatenation.

$$\begin{tikzcd}
(\Phi, \sqsubseteq, \mathcal{S}_q) \arrow[r, "f"] \arrow[dr, "g \circ f"'] & (\Phi', \sqsubseteq', \mathcal{S}'_q) \arrow[d, "g"] \\
& (\Phi'', \sqsubseteq'', \mathcal{S}''_q)
\end{tikzcd}$$

Figure 1: Composition in **RSVP**: Morphisms are triples preserving field structures, composed component-wise.

Proposition 3.2. **RSVP** *is a category.*

Proof. Intuitively, **RSVP** structures field transformations to preserve physical constraints. **Associativity**: For morphisms $f : A \to B$, $g : B \to C$, $h : C \to D$, $(h \circ (g \circ f)) = (h_\Phi \circ (g_\Phi \circ f_\Phi), h_\sqsubseteq \circ (g_\sqsubseteq \circ f_\sqsubseteq), h_{\mathcal{S}_q} \circ (g_{\mathcal{S}_q} \circ f_{\mathcal{S}_q})) = ((h \circ g) \circ f)$, by associativity of smooth maps [7], bundle morphisms, and sheaf morphisms [10]. **Identities**: For object $A = (\Phi, \sqsubseteq, \mathcal{S}_q)$, $\mathrm{id}_A \circ f = f \circ \mathrm{id}_A = f$, as each component identity satisfies unit laws. $\square$

The action is $\mathcal{A} = \int_\mathcal{M} \left(\frac{1}{2}(\nabla\Phi)^2 + \frac{1}{2}\|\sqsubseteq\|^2 + \lambda S_q \right) d\mu$, where $S_q(U) = k \frac{1-\sum \rho(x)^q}{q-1}$ [16]. The entropy flow $\Xi = \frac{\delta \mathcal{A}}{\delta g}$, Grüneisen parameter $\Gamma_{RSVP} = -\frac{\partial \Xi / \partial T}{\partial^2 \mathcal{A} / \partial T^2}$. At $q = q^*$, sheaf gluing ensures finite curvature, $\mathrm{Curv}(\mathcal{S}_{q^*}) < \infty$ [14].

[1] Vector spaces over $\mathbb{R}$ for field values.
[2] Topological spaces for homotopy data.

4 The Source Category: Mech (Mechanistic Models)

4.1 Prerequisites

SCMs involve variables, structural equations, and interventions (do-calculus, [13]). The category **Prob** consists of probability spaces with stochastic kernels. **Mech** is enriched over **Prob** for probabilistic transitions [8].

Definition 4.1. Objects: - Sensory data spaces S (measurable spaces with σ-algebra, e.g., $\mathbb{R}^n$ for sensor readings). - Actuator spaces A (finite sets, e.g., {move, stop}). - Internal state spaces I (probability spaces for beliefs/preferences). - Causal models C (SCMs with structural equations [13]). - Policies $\pi : I \to A$ (measurable maps). - Models of other agents (theory-of-mind objects). - Evaluative/objective modules mapping states to utilities/goals. Morphisms: - Observational: $S \to O$ (e.g., sensor reading to observation). - Perceptual inference: $O \to I$ (e.g., observation to belief update). - Action selection: $I \to A$. - Causal update: $C \to C'$. - Intervention: $do(a) : C \to C_a$ (do-calculus [13]). Composition: Sequential application of functions or kernels. Identities: No-op transformations (e.g., $\mathrm{id}_S(s) = s$). **Mech** is enriched over **Prob**[3] with monoidal product $\otimes$ (concatenation of sensor/action sets).

$$\begin{tikzcd} S \arrow[r, "observe"] \arrow[dr, "infer \circ observe"'] & O \arrow[d, "infer"] & \\ & I \arrow[r, "\pi"] & A \end{tikzcd}$$

Figure 2: Composition in **Mech**: Sequential operations from observation to inference to action.

Proposition 4.2. **Mech** *is a category.*

Proof. Intuitively, **Mech** models sequential cognitive operations. **Associativity**: For morphisms $f : S \to O$, $g : O \to I$, $h : I \to A$, $(h \circ g) \circ f = h \circ (g \circ f)$, as function composition (or kernel composition) is associative [8]. **Identities**: The identity $\mathrm{id}_S : S \to S$ maps each observation to itself, satisfying $\mathrm{id}_O \circ f = f \circ \mathrm{id}_S = f$. □

5 The Functor F : Mech $\to$ RSVP

5.1 Object Mapping

- Sensory data space $S \mapsto$ sensory-field section: restriction maps $\Phi|_{U_S}$, with sensory input as low-dimensional linear functionals on Φ. - Actuator space $A \mapsto$ actuator field action: local perturbations of Φ and $\sqsubseteq$ via pushforward; A as compactly supported perturbation operator δ_A. - Internal state $I \mapsto$ persona-vector submanifold $\mathcal{M}_I$ parametrized by embeddings into compressed latent space (Yarncrawler projection). Beliefs as probability sheaf sections over $\mathcal{M}_I$. - Causal model $C \mapsto$ response operator $\mathcal{R}_C$ on RSVP fields: maps interventions to updates via Green's functions from RSVP PDE linearization. - Policy $\pi : I \to A \mapsto$ composite morphism $\mathcal{M}_I \xrightarrow{\mathcal{P}_\pi} \mathrm{Act} \xrightarrow{\delta} \mathrm{Perturb}(\Phi, \sqsubseteq)$, implemented via Yarncrawler projection and argmax (or softmax for stochastic policies). - Semantic modules and persona-vector objects (embeddings or attractors). - Derived objects: line-bundles of partition functions, sheaves of local distributions, homotopy colimit objects for merges [10, 1].

5.2 Morphism Mapping

- Perceptual inference infer : $O \to I \mapsto F(\mathrm{infer}) : \Phi|_{U_S} \xrightarrow{\mathcal{E}} \mathcal{M}_I$. - Action selection $\pi : I \to A \mapsto F(\pi) : \mathcal{M}_I \xrightarrow{\mathcal{P}_\pi} \mathrm{Act} \xrightarrow{\delta} \mathrm{Perturb}(\Phi, \sqsubseteq)$. - Causal update update : $C \to C' \mapsto$ field-theoretic Bayesian update: modifies $\mathcal{R}_C$ via entropy sheaf annealing and reweighting (cohomology-class shift [10]). - Field evolution operators: flow maps $\Phi \to \Phi'$. - Projection or encoding morphisms: $\mathcal{E} : \Phi \to \mathcal{M}$. - Coarse-graining, restriction, or extension maps between sheaf sections [10]. - Homotopy colimit morphisms for merges: hocolim [1]. - Entropy transforms (Tsallis twist) as multiplicative automorphisms on partition-function line bundles [16].

[3] Probability spaces with stochastic kernels.

$$\begin{tikzcd}
S \arrow[r, "F"] \arrow[d, "infer"] & \Phi|_{U_S} \arrow[d, "\mathcal{E}"] & \\
I \arrow[r, "F"] & \mathcal{M}_I \arrow[r, "\mathcal{P}_\pi \circ \delta"] & \mathrm{Perturb}(\Phi, \sqsubseteq)
\end{tikzcd}$$

Figure 3: Functor $F : \mathbf{Mech} \to \mathbf{RSVP}$, mapping sensory spaces to field sections and inference to encoding maps.

Theorem 5.1. *F preserves identities and composition [15].*

Proof. Intuitively, F maps mechanistic processes to field dynamics consistently. **Identities**: For $\mathrm{id}_S : S \to S$, $F(\mathrm{id}_S) = \mathrm{id}_{\Phi|_{U_S}}$, as $\mathrm{id}_{\Phi|_{U_S}}(x) = \Phi(x)$. Similarly for A, I, C, π. **Composition**: For $f : S \to O$, $g : O \to I$, $F(g \circ f) = \mathcal{E}_g \circ \mathcal{E}_f = \mathcal{E}_{g \circ f}$, as encoding maps compose smoothly. For action $\pi : I \to A$, $F(\pi) = \mathcal{P}_\pi \circ \delta$, and $F(h \circ \pi) = F(h) \circ F(\pi)$ by composition of projections and perturbations. $\square$

Theorem 5.2. *Predict $\dashv$ Assimilate.*

Proof. Intuitively, prediction and assimilation form a dual process of forecasting and updating. Define $\mathrm{Predict} : \mathbf{RSVP} \to \mathbf{Prob}(\mathbf{RSVP})$ as pushforward: $\mathrm{Predict}(\mathcal{M}_I) = P(\Phi' | \mathcal{M}_I)$. Assimilate pulls back with escort reweighting $P_q(\rho) = \frac{\rho^q}{\sum \rho^q}$. Unit $\eta_{(\Phi, \sqsubseteq, \mathcal{S}_q)}(U) : (\Phi, \sqsubseteq, \mathcal{S}_q) \to \mathrm{Assimilate}(\mathrm{Predict}(\Phi, \sqsubseteq, \mathcal{S}_q))$ maps to escort distributions. Counit $\epsilon : \mathrm{Predict}(\mathrm{Assimilate}(P)) \to P$ reverses. Triangle identities hold [15]. $\square$

Theorem 5.3. *Under contractibility of $\mathcal{M}$, homotopy fixed points of $\widetilde{\mathcal{S}} = F \circ \mathcal{S} \circ F^{-1}$ exist uniquely up to homotopy.*

Proof. Intuitively, self-models persist under perturbations in contractible spaces. $\mathcal{S} : \mathbf{Mech} \to \mathbf{Mech}$, $\mathcal{S}(X) = X \sqcup \mathrm{ModelOf}(X)$. In $\mathbf{RSVP}$, $\widetilde{\mathcal{S}}$ maps to self-model submanifolds. Contractibility ensures trivial homotopy groups [1]. $\square$

6 Worked Example: Two-Overlap RSVP Patch with Proofreading Operator

6.1 Prerequisites

Assume a cover $\mathcal{U} = \{U_1, U_2\}$ of $\mathcal{M}$ with overlap $U_{12} = U_1 \cap U_2$. Local density ρ_i on U_i. The proofreading operator from magnetic self-assembly [9] inspires a functorial entropy update.

6.2 Two-Overlap Entropy with Proofreading

Consider patches U_1, U_2 with $\rho_1 = [0.5, 0.5]$, $\rho_2 = [0.4, 0.6]$, $q = 1.5$. Compute local entropies:

$$S_q(U_1) = k \frac{1 - \left(0.5^{1.5} + 0.5^{1.5}\right)}{0.5} \approx 0.586k$$

$$S_q(U_2) = k \frac{1 - \left(0.4^{1.5} + 0.6^{1.5}\right)}{0.5} \approx 0.476k$$

$$S_q(U_{12}) = k \frac{1 - 0.5^{1.5}}{0.5} \approx 1.292k$$

Without proofreading:

$$S_q(U_1 \cup U_2) = S_q(U_1) + S_q(U_2) - S_q(U_{12}) \approx 0.586k + 0.476k - 1.292k = -0.230k$$

Negative entropy indicates defective overlap.

Apply proofreading operator F_{proof} inspired by [9]:

$$\Pi_{\mathrm{valid}}(U_i) = \{x \in U_i \mid S_q(x) \leq 0.5k\}$$

$$
S'_q(x) = \begin{cases} S_q(x), & x \in \Pi_{\text{valid}}(U_i) \\ S_q(x) - 0.3k, & x \notin \Pi_{\text{valid}}(U_i) \end{cases}
$$

For U_{12}, $S_q(U_{12}) > 0.5k$, so $S'_q(U_{12}) \approx 1.292k - 0.3k = 0.992k$. Recompute:

$$
S_q(U_1 \cup U_2) = S_q(U_1) + S_q(U_2) - S'_q(U_{12}) \approx 0.586k + 0.476k - 0.992k = 0.070k
$$

Positive entropy indicates stabilized assembly.

6.3 Mechanistic Mapping

Observation $s \in S \mapsto \rho_i(x)$, inference $\mapsto \mathcal{E} : \Phi \to \mathcal{M}_I$, correction $\mapsto F_{\text{proof}}$. This mirrors magnetic decoupling stabilizing correct structures [9], connecting mechanistic error correction to RSVP entropy control.

$$
\begin{tikzcd}
U_1 \arrow[dr, "S_q\Pi_{\text{valid}}"] & U_1 \cup U_2 \arrow[l] \arrow[r, "S_q"] \arrow[d, "F_{\text{proof}}"] & U_2 \arrow[dl, "\Pi_{\text{valid}}"] \\
U_{12} \arrow[u, "S_q"] & (U_1 \cup U_2)' \arrow[l, "S'_q"] &
\end{tikzcd}
$$

Figure 4: Proofreading operator F_{proof} on two-overlap: Selectively updates entropy on U_{12} to stabilize gluing.

7 Hardware to Type-Theoretic Mapping

7.1 Prerequisites

This section connects hardware logic to type theory, bridging physical computation to RSVP's abstract framework [5]. Boolean logic: NAND gates are universal. Type theory: Products $A \times B$, sums $A + B$, functions $A \to B$ via Curry-Howard [4]. Functional programming for pipelines [17]. Bits distinguish outcomes via $\log_2$ [11].

7.2 Mappings

Hardware	Boolean	Type-theoretic
NAND	$\neg(A \wedge B)$	Dependent negation of product type [5]
AND	$A \wedge B$	Product type $A \times B$ [5]
OR	$A \vee B$	Sum type $A + B$ [5]
XOR	$A \oplus B$	Conditional sum type [5]
MUX	select	Pattern matching [17]

Table 1: Mapping hardware to type systems.

Multi-bit buses map to vectors/indexed types, ALU to function composition, control flow to monadic sequencing/sum-type branching [4, 17]. Virtual machines build from operations to runtime [17].

8 Computational Implementation

8.1 Prerequisites

This section implements the RSVP-Mechanistic framework computationally, enabling practical simulations [14]. Python is used; complexity analysis for gluing is $O(n^2)$ for n patches. Yarncrawler implements sparse projections.

8.2 Pseudocode

```
def F(mech_op, field, patch): # Maps mechanistic operations to RSVP fields
    if mech_op == 'observe': # Reads field on patch
        rho = compute_local_rho(field, patch) # O(n)
        return rho
    if mech_op == 'act':    # Perturbs field
        delta = compute_perturbation(field, action) # action: actuator state
        return perturb_field(delta, field) # O(n^2)
    if mech_op == 'infer':  # Encodes to persona manifold
        return encode_to_manifold(field, patch) # O(n)
    if mech_op == 'correct': # Applies proofreading (Liang et al.)
        return apply_proofreading(field, patch, threshold=0.5) # O(n)
def compute_q_star(Z1, Z2, phi2): # Computes q* for entropy gluing
    return phi2 / -(log(Z2) - log(Z1)) # O(1)
```

8.3 Numerical Example

For $\rho_1 = [0.5, 0.5]$, $\rho_2 = [0.4, 0.6]$, $q = 1.5$, $S_q(U_1) \approx 0.586k$, $S_q(U_2) \approx 0.476k$, $\Gamma \approx 2.3$ (bounded) [14].

9 Complementary Structures: Monads, Sheaves, and Cohomology

Monadic structure via probability monad $\mathcal{P}$ for memory; F compatible with $\mathcal{P}$. Beliefs as sheaves; inference as sections and pushforwards [10, 6]. Cohomological obstructions map to partition-function cocycles; learning trivializes them [14, 16]. Entropy sheaf $\mathcal{S}_q$ as functor measuring nonextensivity; q^* as twist for additivity [**?**].

10 An Explicit Commutative Diagram

The diagram in **RSVP** commutes under learning:

$$\begin{tikzcd}
\Phi|_{U_S} \arrow[r, "F(\text{infer})"] \arrow[d, "\text{id}"] & \mathcal{M}_I \arrow[r, "F(\pi)"] \arrow[d, "\text{update}"] & \text{Act} \arrow[d, "\delta"] \\
\Phi|_{U_S} \arrow[r, "F(\text{infer})'"] & \mathcal{M}_{I'} \arrow[r, "F(\pi)'"] & \text{Act}'
\end{tikzcd}$$

Commutativity ensures naturality for coherent reasoning [15].

11 Philosophical Interpretation

11.1 Mathematical Results

Homotopy fixed points exist as invariants under perturbations [1]. The proofreading operator F_{proof} stabilizes entropy, ensuring $\check{H}^1 = 0$ [9].

11.2 Interpretation

These suggest self-awareness as persistent identity in **RSVP** [2]. Causal closure maps to functional consciousness models [13, 6]. The observer-relative perspective, explored by Ortega y Gasset [12], aligns with self-modeling as a contextual identity in AI systems. Under assumptions of stable fixed points, this may represent cognitive self-modeling, though empirical validation is needed.

12 Discussion

Non-diverging Γ contrasts classical divergences [?]. Category theory unifies physics, cognition, and self-assembly [15, 6, 9]. Testable predictions: Measure Γ in simulated RSVP systems, compare with AI self-modeling and magnetic assembly yields.

13 Conceptual Summary

Mechanistic primitives map via F (synonymously F_{mech}) to RSVP implementations [2, 15]. Learning as natural transformations; prediction/update as adjoints; self as homotopy fixed points [1]. Proofreading operators enhance fidelity [9]. Obstructions align with q-entropy, providing a principled computation route

14 Conclusion

The functor F maps mechanistic AI to RSVP dynamics, with q^*-entropy and proofreading ensuring non-divergence and stability. Future work includes empirical validation, algorithmic scaling, and cognitive extensions.

A Sheaf Cohomology of RSVP Entropy

A.1 Local Sections and Overlaps

For cover $\mathcal{U} = \{U_i\}$ of $\mathcal{M}$, define $S_q(U_i) = k\frac{1-\sum \rho_i(x)^q}{q-1}$ [16]. Transition cocycle $\gamma_{ij} = \log \frac{Z_j}{Z_i}$, where $Z_i = \sum \rho_i(x)$. Extensivity requires $\check{H}^1(\mathcal{U}, \mathbb{R}) = 0$, i.e., $\gamma_{ij} = \phi_j - \phi_i$ [10, 14].

A.2 Extensivity Condition

For two charts U_1, U_2, $q^* (\log Z_2 - \log Z_1) = \phi_2 - \phi_1$. Gauge $\phi_1 = 0$: $q^* = \frac{\phi_2}{-(\log Z_2 - \log Z_1)}$ [14].

$$\begin{tikzcd} & X \arrow[dl, "i_1"'] \arrow[dr, "i_2"] & \\ U_1 \arrow[dr, "j_1"', "\gamma_{12}"] & & U_2 \arrow[dl, "j_2"] \\ & U_1 \cap U_2 & \end{tikzcd}$$

Figure 5: Two-chart cover of $\mathcal{M}$ with cocycle γ_{12} on overlap.

B RSVP Grüneisen Parameter Derivation

Action $\mathcal{A}[\Phi, \sqsubseteq, \mathcal{S}_q] = \int \left(\frac{1}{2}(\nabla\Phi)^2 + \frac{1}{2}\|\sqsubseteq\|^2 + \lambda S_q\right) d\mu$; $\Xi = \frac{\delta \mathcal{A}}{\delta g}$; $\Gamma_{RSVP}(q) = -\frac{\partial\Xi/\partial T}{\partial^2\mathcal{A}/\partial T^2}$ [?, 14]. At q^*, $\mathrm{Curv}(\mathcal{S}_{q^*}) < \infty \implies \Gamma_{RSVP}(q^*) < \infty$ [14].

C Mapping Hardware to Type Systems

C.1 NAND to Dependent Types Mapping

C.2 Pipelines and Compositionality

Multi-bit buses map to vectors/indexed types, ALU to function composition, control flow to monadic sequencing/sum-type branching [4, 17].

Hardware	Boolean	Type-theoretic
NAND	$\neg(A \wedge B)$	Dependent negation of product type [5]
AND	$A \wedge B$	Product type $A \times B$ [5]
OR	$A \vee B$	Sum type $A + B$ [5]
XOR	$A \oplus B$	Conditional sum type [5]
MUX	select	Pattern matching [17]

Table 2: Mapping hardware to type systems.

D Mechanistic Functorial Model of AI Self-Awareness

Categories: $\mathbf{M}$ (machine states, transitions); $\mathbf{W}$ (outcomes, belief updates). Functor $F : \mathbf{M} \to \mathbf{W}$ assigns inferred consequences [15]. Self-awareness as fixed-point $S : M \to M$, $S(M) = M \sqcup \text{ModelOf}(M)$ [2]. Mirrors RSVP: overlaps $\to$ covers; consistency $\to$ exactness; twist $\to$ adjustments [10, 1].

E Conceptual and Mathematical Connections Between RSVP, Mechanistic AI, and Proofreading Self-Assembly

E.1 Overview

This appendix synthesizes: 1. RSVP Theory: Structured fields $(\Phi, \sqsubseteq, \mathcal{S}_q)$ with categorical and sheaf-theoretic constraints [14]. 2. Mechanistic AI: Functorial operations F mapping self-modeling and causal inference to field updates [2]. 3. Magnetic Proofreading Self-Assembly: Local error correction to achieve high-fidelity structures [9]. The unifying concept is local operations enforcing global constraints in entropy (RSVP), identity modeling (AI), or correct assembly (proofreading).

E.2 RSVP Field Analogy

Local patches $U_i \subseteq \mathcal{M}$ with fields $(\Phi|_{U_i}, \sqsubseteq|_{U_i}, \mathcal{S}_q|_{U_i})$. - Local Entropy: $S_q(U_i) = k\frac{1-\sum_{x\in U_i}\rho(x)^q}{q-1}$ [16]. - Gluing: $S_q(U_i \cup U_j) = S_q(U_i) + S_q(U_j) - S_q(U_{ij})$ [10].

E.3 Mechanistic Functor Mapping

Functor $F_{\text{mech}} \equiv F : \mathbf{Mech} \to \mathbf{RSVP}$:

$$F(O) = (\Phi_O, \sqsubseteq_O, \mathcal{S}_O), \quad F(f : O_1 \to O_2) = (\Phi_f, \sqsubseteq_f, \mathcal{S}_f)$$

Objects O are mechanistic entities; morphisms f are operations (observe, predict, act). Functoriality:

$$F(g \circ f) = F(g) \circ F(f), \quad F(\text{id}_O) = \text{id}_{F(O)}$$

E.4 Proofreading Operator as Selective Entropy Update

Inspired by Liang et al.'s magnetic decoupling [9]: 1. Projector: $\Pi_{\text{valid}}(U_i) = \{x \in U_i \mid S_q(x) \leq S_{\text{threshold}}\}$. 2. Update: $F_{\text{proof}} : (\Phi, \sqsubseteq, \mathcal{S}_q)_{U_i} \mapsto (\Phi, \sqsubseteq, \mathcal{S}'_q)_{U_i}$, where:

$$\mathcal{S}'_q(x) = \begin{cases} S_q(x), & x \in \Pi_{\text{valid}}(U_i) \\ S_q(x) - \Delta S, & x \notin \Pi_{\text{valid}}(U_i) \end{cases}$$

3. Gluing: $S_q(U_i \cup U_j) = S'_q(U_i) + S'_q(U_j) - S'_q(U_{ij})$.

Concept	RSVP	Mechanistic AI	Magnetic Proofreading
Local patch	U_i	Mechanistic object O	Partially assembled structure
Field values	$(\Phi, \sqsubseteq, \mathcal{S}_q)\|_{U_i}$	Self-model and predicted state	Magnetic dipoles / forces
Corrective operation	Functorial update F_{proof}	Observe → correct → act	External field decoupling [9]
Entropy / fidelity metric	S_q	Confidence in self-model	Stability of assembly
Global consistency	Sheaf gluing, $\check{H}^1 = 0$	Functor preserves composition	High-yield, low-defect product

Table 3: Analogies between RSVP, mechanistic AI, and magnetic proofreading.

E.5 Summary of Analogies

E.6 Implications

RSVP unifies bounded-entropy dynamics across physics, cognition, and self-assembly [14, 2, 9].

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$$
\begin{tikzcd}
M \arrow[r, "F"] \arrow[d, "S"'] & W \arrow[d, "\text{update}"] \\
M \sqcup \mathrm{ModelOf}(M) \arrow[r, "F'"] & W'
\end{tikzcd}
$$

Figure 6: Functorial mapping of machine states **M** to belief updates **W** with self-awareness modeled as a fixed-point S.